Quantifying Longitudinal and Lateral Movements of Vehicles

Motorcycles' naturalistic driving behavior

1. Approach-aligned coordinates, kinematics and movement detection
2. Application: motorcycle lane filtering analysis

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Why Align Trajectories to the Approach?

- ✓ Raw projected (UTM) coordinates mix forward and sideways motion
- ✓ An approach frame separates:
  - ✓ **Longitudinal behavior:** stopping, discharge, progress
  - ✓ **Lateral behavior:** filtering, swerving, lane changes
- ✓ Goal: variables that feed the event definition and calibration criteria for lane filtering

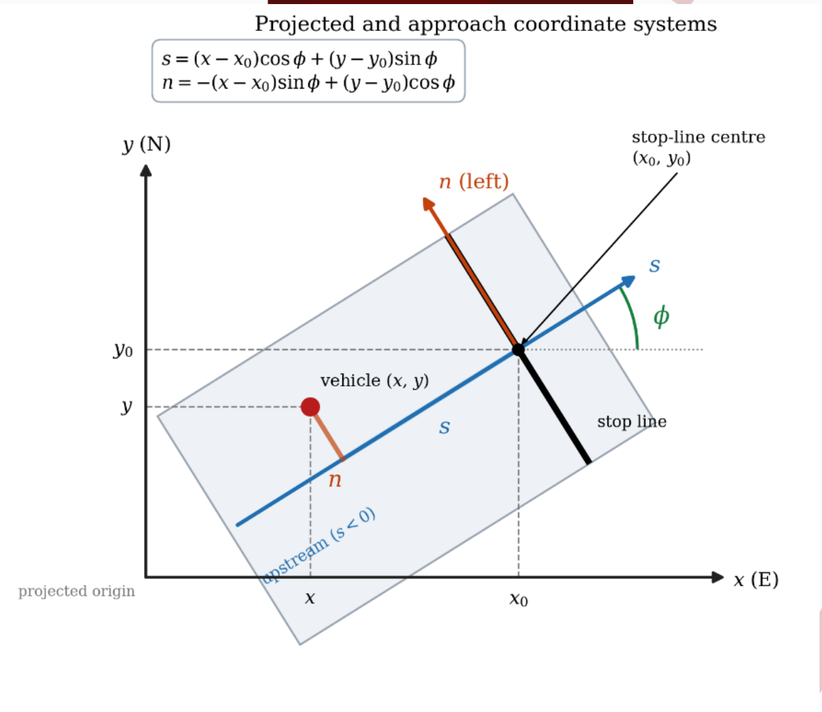


Figure: Conceptual illustration of projected and approach coordinate system

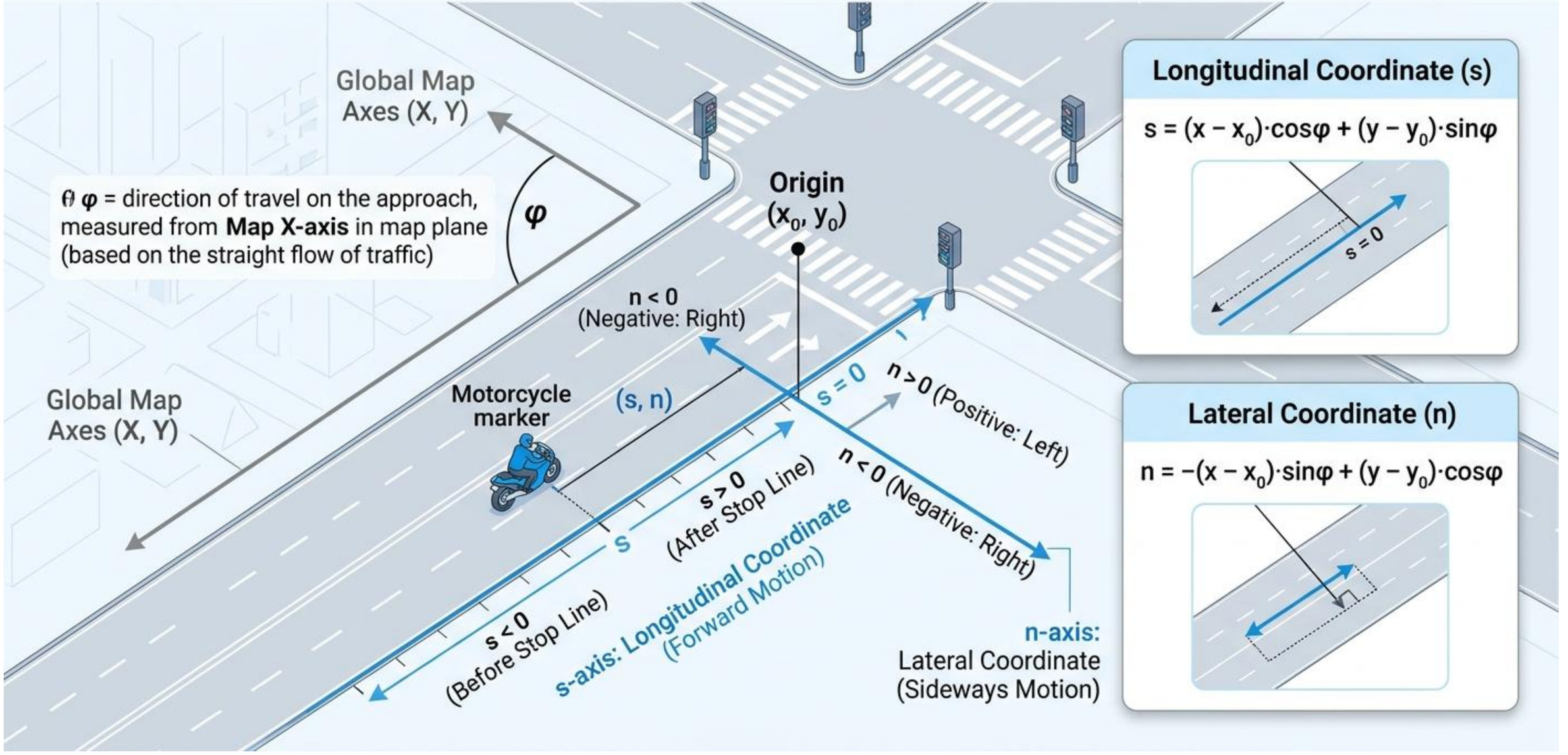
Approach-Aligned Coordinates

- Origin: stop-line center (x_0, y_0); approach azimuth φ
- Transformation:
  $s = (x - x_0)\cos\varphi + (y - y_0)\sin\varphi$
  $n = -(x - x_0)\sin\varphi + (y - y_0)\cos\varphi$
- s: longitudinal distance from the stop line (s = 0). Negative is upstream of the stop line, positive is downstream.
- n: lateral offset from the reference line. Positive is left, negative is right of travel.

Table: Relevant notation for longitudinal and lateral movement

Symbol	Meaning
x, y	Projected (UTM) position of the vehicle reference point (m)
x_o, y_o	Stop-line center in the same projected frame (m)
ϕ	Approach azimuth: direction of travel on the approach, measured in the projected plane (rad)
s, n	Longitudinal (along approach) and lateral (positive to the left of travel) coordinates (m)
v_x, v_y	Smoothed velocity components in the projected frame (m/s)
v_s, v_n	Longitudinal and lateral speed in the approach frame (m/s)
θ	Heading in the projected frame (rad)
ψ	Heading relative to the approach direction (rad)
ω	Heading rate $d\psi/dt$ (rad/s)

Approach-Aligned Coordinate System



Longitudinal Coordinate (s)

$$s = (x - x_0)\cdot\cos\varphi + (y - y_0)\cdot\sin\varphi$$

Lateral Coordinate (n)

$$n = -(x - x_0)\cdot\sin\varphi + (y - y_0)\cdot\cos\varphi$$

Data Used

30 Hz Kalman-smoothed trajectories are used for all computations ($\Delta t \approx 0.0334$ s)

1 s resampled data is used for graphical display only: Automated quality control (QC) and packaging

Fields kept per track: x, y, time, vehicle type

Differentiation amplifies noise, so derivatives are computed only after smoothing

DATA PREPARATION AND SMOOTHING

INPUT: VEHICLE TRACKS



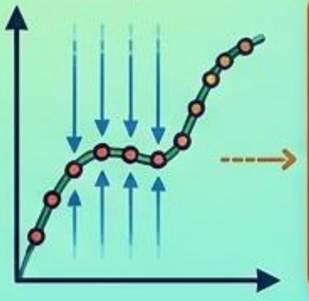
KALMAN SMOOTHING



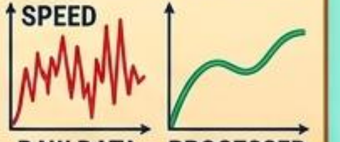


KALMAN FILTER SMOOTHS TRACKS AT 30 Hz

TIME STEP $\Delta t \approx 0.0334$ s



REASON: DIFFERENTIATION MAGNIFIED NOISE

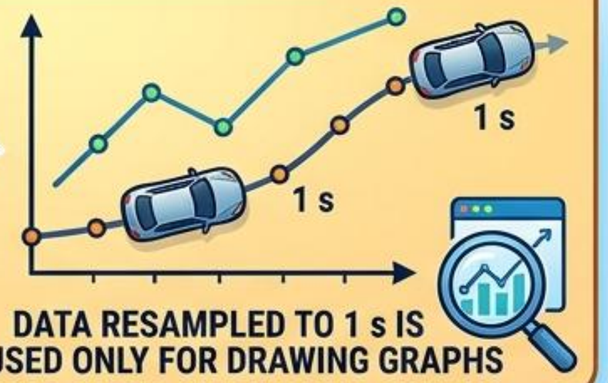


OUTPUT 1: 30 Hz DATA FOR ANALYSIS



30 Hz SMOOTHED DATA IS USED FOR ALL CALCULATIONS

OUTPUT 2: 1 s DATA FOR PLOTS



DATA RESAMPLED TO 1 s IS USED ONLY FOR DRAWING GRAPHS

Speed, Acceleration and Heading

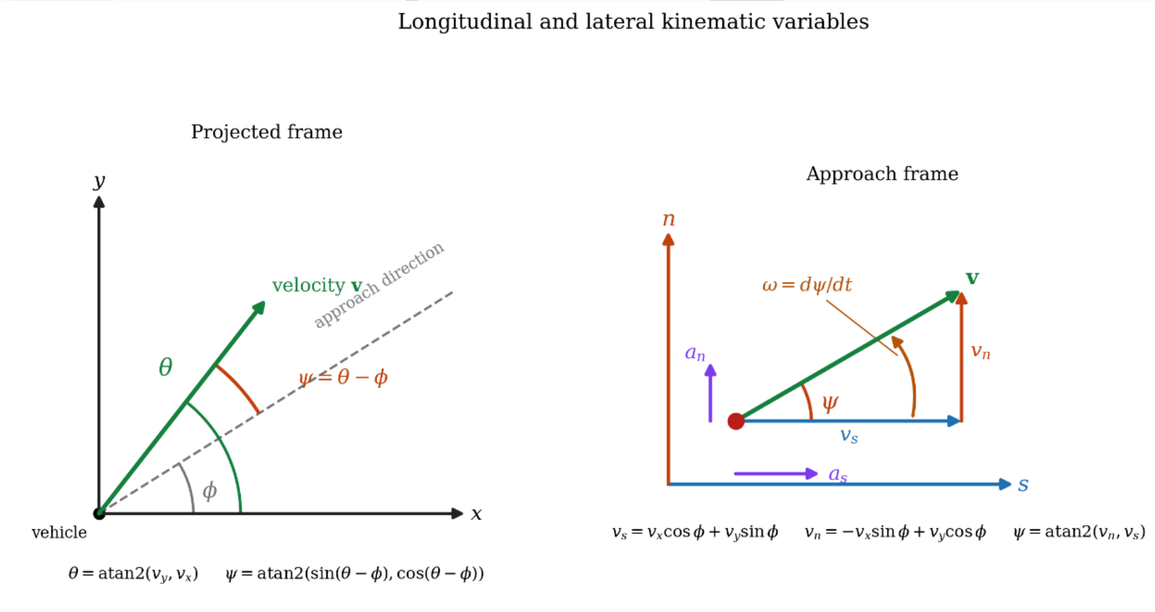


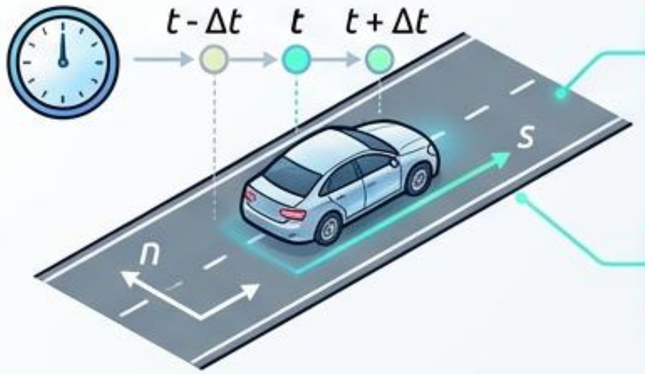
Figure: Conceptual illustration of longitudinal and lateral kinematic variables

- Central difference for velocity:
  - $v_s = [s(t + \Delta t) - s(t - \Delta t)]/2\Delta t$
  - $v_n = [n(t + \Delta t) - n(t - \Delta t)]/2\Delta t$
- Second difference for acceleration:
  - $a_n = [n(t + \Delta t) - 2n(t) + n(t - \Delta t)]/\Delta t^2$
  - a_s is computed the same way from s
- Heading: $\theta = atan2(v_y, v_x)$ in the projected frame, $\psi = atan2(v_n, v_s)$ relative to the approach
- Heading rate: $\omega = d\psi/dt$
- ψ is undefined below 1 km/h because direction is unreliable near zero speed
- Cross-check: compare a_n with the DFS lateral acceleration

SPEED, ACCELERATION & HEADING: MOTION VARIABLES FROM ROAD-ALIGNED POSITIONS

SMOOTHED ROAD-ALIGNED POSITIONS $s(t)$, $n(t)$

LONGITUDINAL & LATERAL SPEED (v_s, v_n)



$$v_s = \frac{s(t+\Delta t) - s(t-\Delta t)}{2\Delta t}$$

LONGITUDINAL SPEED
(CENTRAL DIFFERENCE)

$$v_n = \frac{n(t+\Delta t) - n(t-\Delta t)}{2\Delta t}$$

LATERAL SPEED

LATERAL & LONGITUDINAL ACCELERTATION (a_n, a_s)

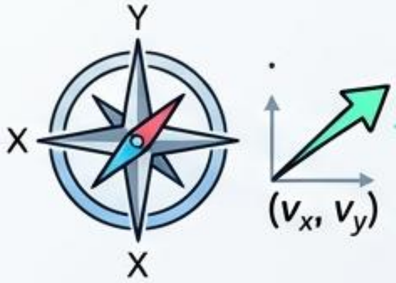
$$a_n = \frac{n(t+\Delta t) - 2n(t) + n(t-\Delta t)}{\Delta t^2}$$

LATERAL ACCELERATION
(SECOND DIFFERENCE)

CROSS-CHECK AGAINST INDEPENDENT ESTIMATE

a_s IS COMPUTED SIMILARLY FROM s

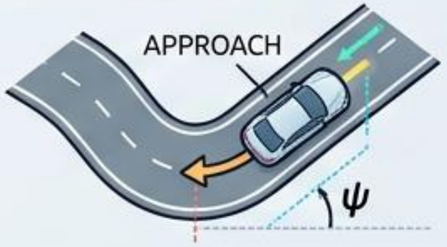
HEADING DEFINITIONS



HEADING IN MAP FRAME (θ)

$$\theta = \text{atan2}(v_y, v_x)$$

ANGLE RELATIVE TO GLOBAL FRAME



HEADING RELATIVE TO APPROACH (ψ)

$$\psi = \text{atan2}(v_n, v_s)$$

ANGLE RELATIVE TO ROAD DIRECTION
POSITIVE ψ MEANS VEHICLE POINTS LEFT



HEADING UNDEFINED BELOW 1 KM/H

DIRECTION UNRELIABLE NEAR ZERO SPEED

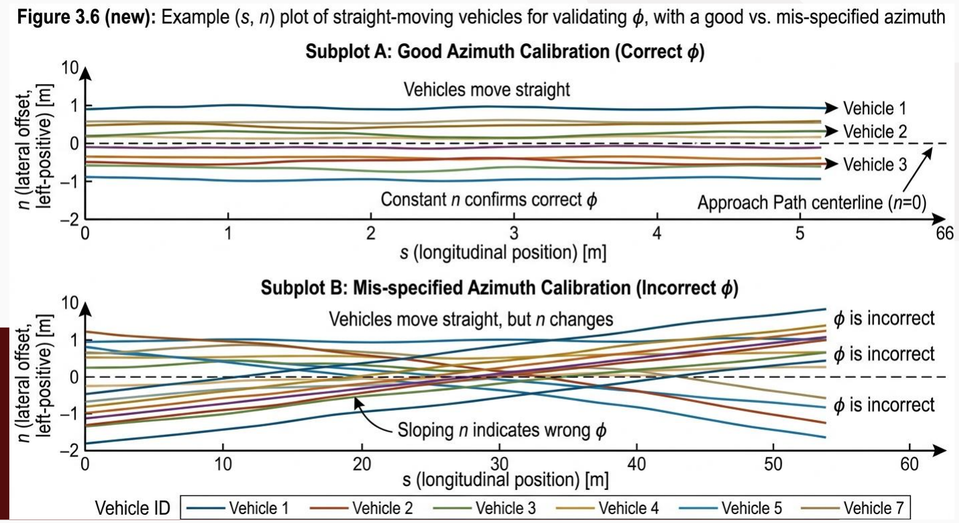


HEADING RATE (ω)

$$\omega = \frac{d\psi}{dt}$$

HOW **FAST** THE VEHICLE IS TURNING (ANGULAR VELOCITY)

Workflow Overview (Steps 1–4)

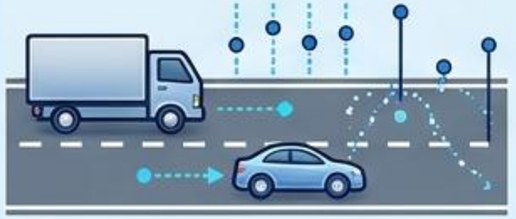


- 1. Load smoothed 30 Hz Kalman trajectories
- 2. Define the approach frame (origin, φ, left-positive n)
- 3. Transform (x, y) to (s, n)
- 4. Validate: plot (s, n). Straight-moving vehicles should keep constant n; otherwise φ is wrong and must be corrected.

COMPUTATION WORKFLOW: Smoothed Trajectories to Longitudinal & Lateral Dataset

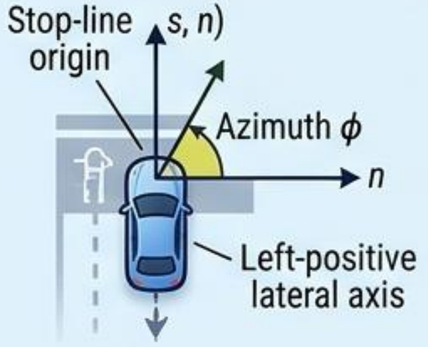
1. PREPARE

1. LOAD TRACKS



Load the smoothed 30 Hz tracks

2. DEFINE APPROACH FRAME



2. TRANSFORM AND CHECK

1. CONVERT COORDINATES



Convert (x, y) to (s, n)

CHECK: CONSTANT *n*?

NO **NO** **YES**

CORRECT *ϕ*

Redoin frame definition

3. DERIVE MOTION

1. COMPUTE ABSOLUTE MOTION



CONTINUE

- Speeds (**v**)
- Headings (ψ)
- Accelerations (**a**)
- Accelerations (**a**)
- Heading rate ($\dot{\psi}$)

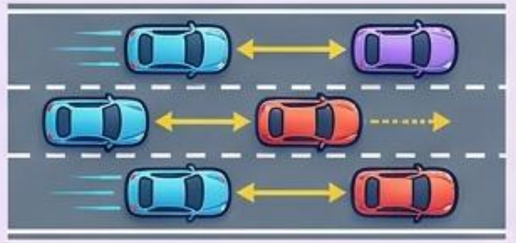
4. DETECT AND COMPARE

1. DETECT MOVEMENT



Detect longitudinal and lateral movement

2. COMPUTE RELATIVE QUANTITIES



Compute relative quantities to neighboring vehicles

5. EXPORT

1. WRITE OUTPUT TABLE



Write the output table



PREPARE | TRANSFORM AND CHECK | DERIVE MOTION | DETECT AND COMPARE | EXPORT

Workflow Overview (Steps 5–6)

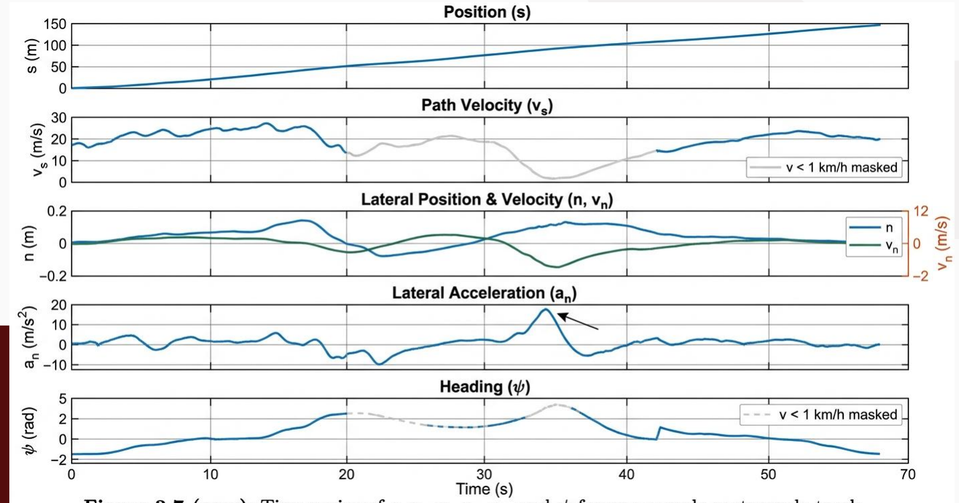
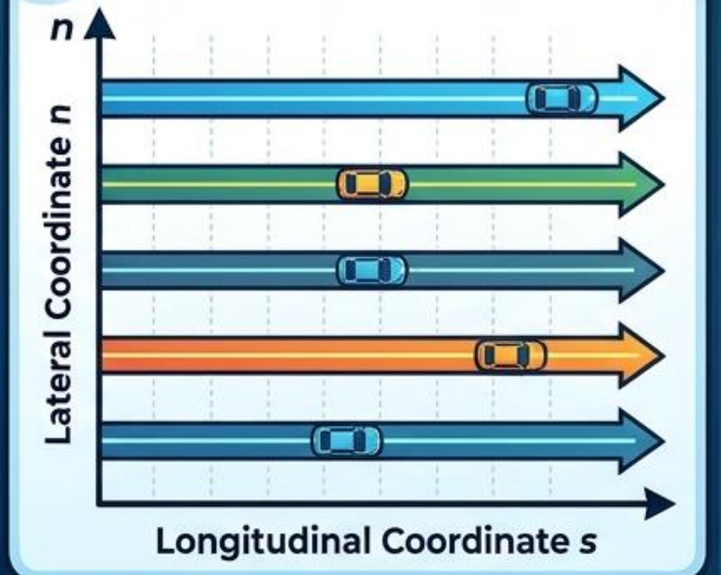


Figure 3.7 (new): Time series of s, v_s, n, v_n, a_n and ψ for one sample motorcycle track.

- 5. Compute v_s, v_n, θ and ψ; mask heading below 1 km/h
- 6. Compute a_s, a_n and ω; cross-check against DFS acceleration

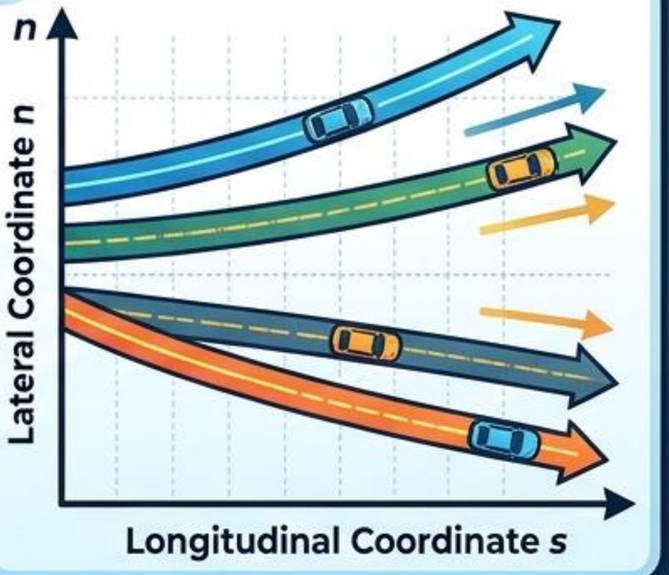
Validating the Coordinate Transform

CORRECT ϕ: No Drift



Tracks stay flat at constant *n*.
ϕ is aligned with road direction.

WRONG ϕ: Lateral Drift



Tracks drift sideways, tilting up or down. Incorrect ϕ estimation.

THE FIX

RE-ESTIMATE ϕ FROM:

1. Lane Markings
or
2. Dominant Traffic Flow.

Detecting Longitudinal and Lateral Movement (Step 7)

- **Longitudinal movement:** $v_s$and $a_s$describe stopping, discharge and progress. Progress is $\Delta s = s(t_2) - s(t_1)$.
- **Lateral movement starts when all three hold:**
  - $|v_n| > \varepsilon_n$(lateral speed threshold)
  - $|\psi| > \varepsilon_\psi$(heading threshold)
  - condition persists for at least T_{lat} Record the net lateral displacement Δn and whether the vehicle crosses a lane boundary or settles in a lane
- Sign of ψ gives direction: positive is left, negative is right

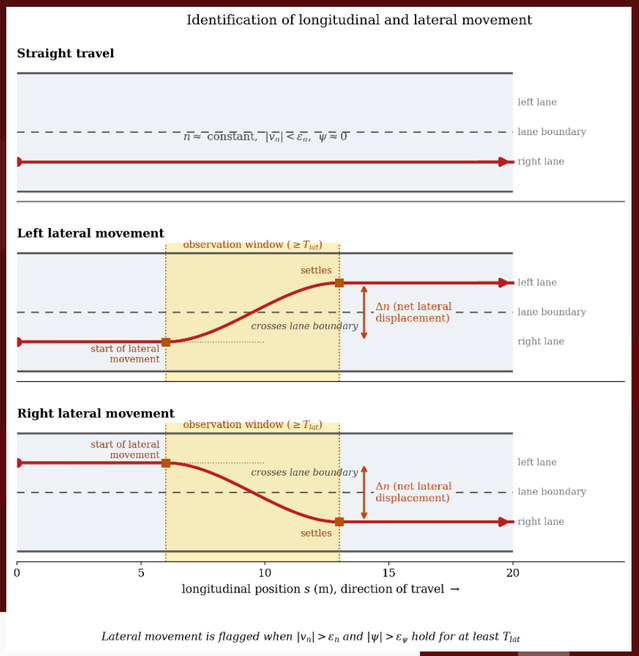
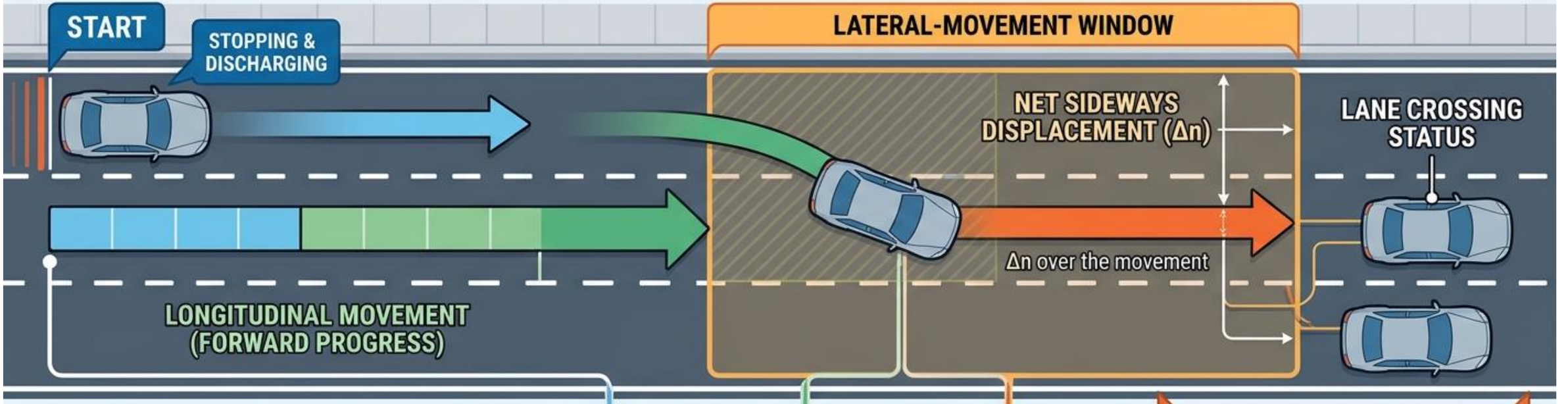


Figure: Conceptual illustration of longitudinal and lateral movement identification

DETECTING LONGITUDINAL AND LATERAL MOVEMENT

Rules for recognizing **forward progress** and **sustained sideways movement**



Key Metrics

Speed (v_s) & Acceleration (a_s) describe stopping, discharging, and forward motion.

Progress between two times is $\Delta s = s(t_2) - s(t_1)$

GATE 1 (SPEED)

LATERAL SPEED: $|v_n|$ > Threshold

GATE 2 (HEADING)

HEADING: $|\psi|$ > Threshold

GATE 3 (DURATION)

LATERAL MOVEMENT starts when all three conditions hold together.

RECORDED OUTCOMES

DIRECTION

SUSTAINED +ψ means Movement to the **LEFT**

SUSTAINED -ψ means Movement to the **RIGHT**

Step 7 (Contd.)

Table: Detection Thresholds for Lateral Vehicle Movement

Parameter	Symbol	Value	Calibration Basis & Justification
Lateral Speed Threshold	ε_n	0.15 m/s	Calibrated against trajectory noise floors to filter out high-frequency GPS jitter, sensor noise, and minor lane-keeping oscillations while remaining sensitive to genuine lateral transitions.
Heading Angle Threshold	ε_ψ	0.02 rad$(\approx 1.15^\circ)$	Determined by analyzing standard highway alignment curvature and normal driver micro-steering variance, ensuring minor steering corrections do not trigger false lane changes.
Time Persistence Threshold	T_{lat}	0.50 s	Based on temporal filtering requirements to reject transient disturbances (such as avoiding potholes, debris, or brief crosswinds) and verify deliberate maneuver intent.

Interaction with Neighboring Vehicles (Step 8)

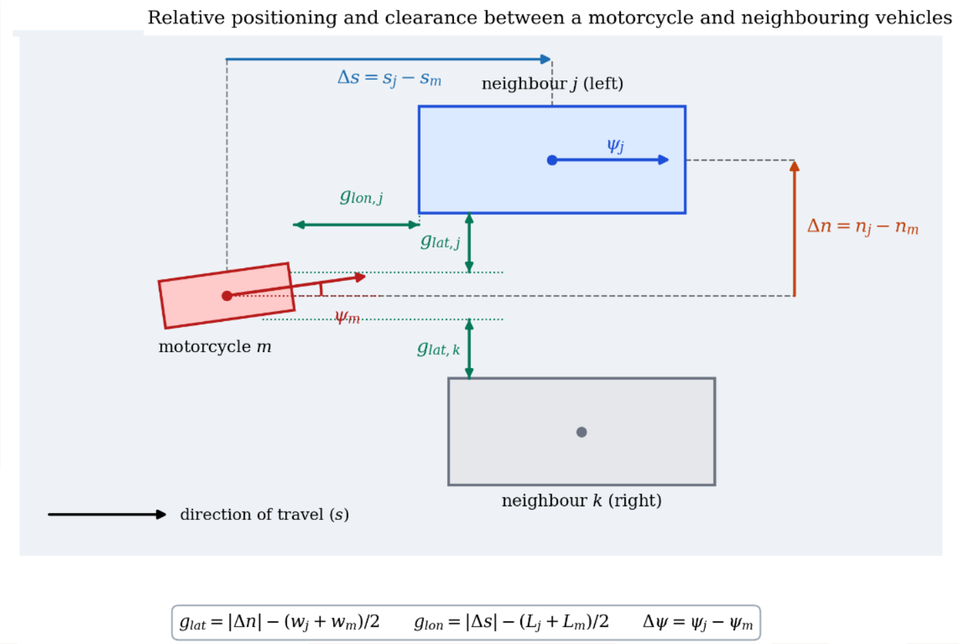


Figure: Conceptual illustration of relative positioning and clearance between motorcycles and neighboring vehicles

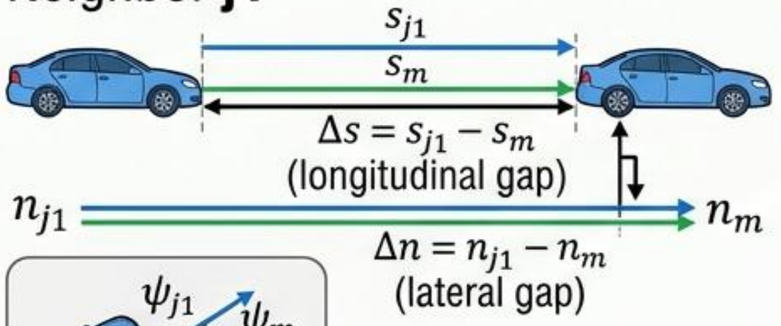
- Relative quantities between vehicle j and motorcycle m:

$$\Delta s = s_j - s_m$$
$$\Delta n = n_j - n_m$$
$$\Delta\psi = \psi_j - \psi_m$$

- Edge-to-edge clearance uses standard vehicle dimensions
- $\Delta\psi \approx 0$ means parallel travel; large $\Delta\psi$ means crossing or merging

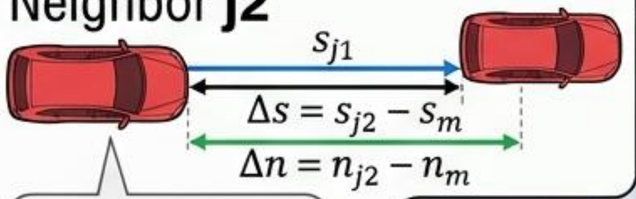
Relative Positioning to Neighbors

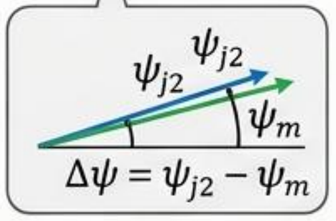
Neighbor j1



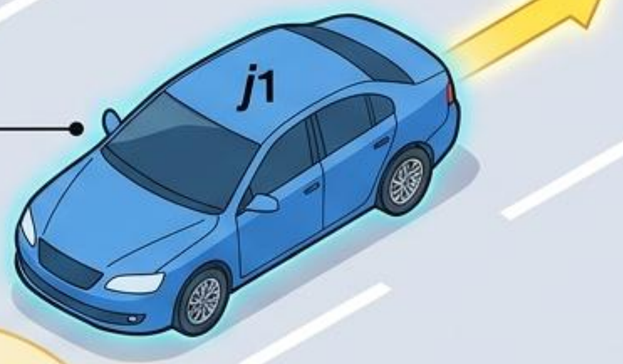
$\Delta\psi$ near zero means the two travel parallel

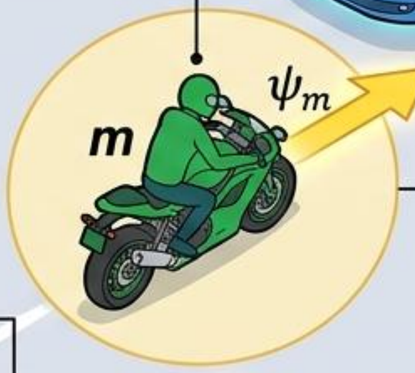
Neighbor j2



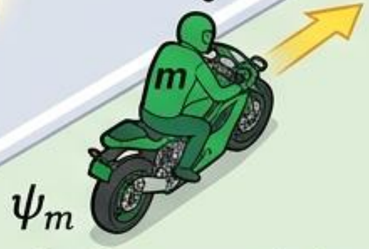


A large $\Delta\psi$ means one is crossing or merging



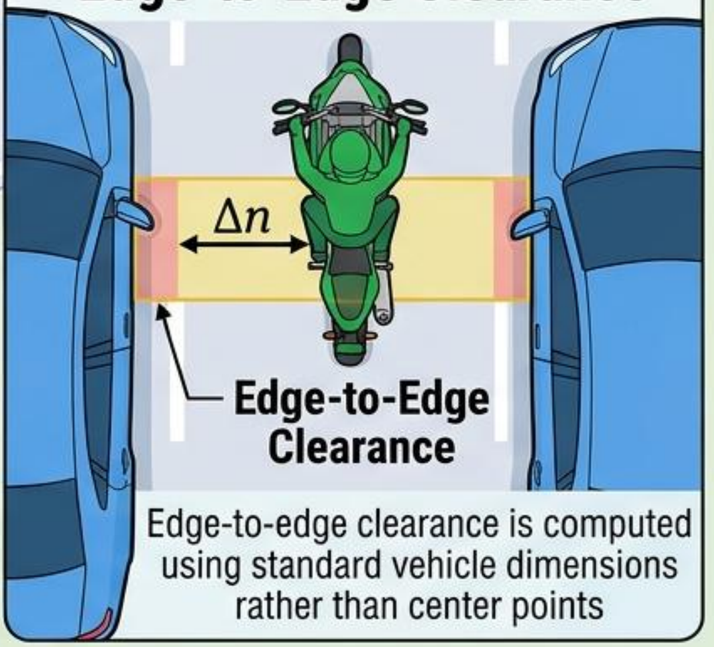






Each motorcycle has its own heading, which changes when it swerves

Edge-to-Edge Clearance



Edge-to-edge clearance is computed using standard vehicle dimensions rather than center points

Output Table (Step 9)

- One record per track and time step
- Columns: Track ID, type, t, x, y, s, n, v_s, v_n, a_s, a_n, θ, ψ, ω
- θ, ψ and ω are included so the filtering and lane-change rules can use them directly

Table: Sample rows of the output table for a motorcycle track

Track ID	type	t [s]	x [m]	y [m]	s [m]	n [m]	v_s[m/s]	v_n[m/s]	a_s[m/s²]	a_n[m/s²]	θ [rad]	ψ [rad]	ω [rad/s]
101	motorcycle	0.0	12.45	4.12	12.50	0.05	11.20	0.02	0.45	0.01	0.12	0.00	0.01
101	motorcycle	0.1	13.58	4.13	13.62	0.06	11.25	0.01	0.50	-0.02	0.12	0.01	0.02
101	motorcycle	0.2	14.71	4.15	14.75	0.05	11.31	-0.01	0.52	0.00	0.13	0.01	-0.01
101	motorcycle	0.3	15.85	4.17	15.89	0.04	11.37	-0.03	0.48	0.03	0.11	0.00	-0.03
101	motorcycle	0.4	16.99	4.20	17.03	0.04	11.41	0.00	0.42	0.01	0.12	0.00	0.00

OUTPUT DATASET: STRUCTURED VEHICLE TRAJECTORY DATA FOR FILTERING AND LANE-CHANGE RULES

Final table with one row per vehicle per time step. Designed for direct use by analysis rules without further calculations.

IDENTITY			POSITION				MOTION				DIRECTION		
Track ID	Vehicle Type	Time t	Map Position x	Map Position y	Road-aligned Position s	Road-aligned Position n	Speed v_s	Speed v_n	Acceleration a_s	Acceleration a_n	Heading θ	Heading ψ	Heading Rate ω
001	Goat	00:02	37.9	20.0	0.8	0.1	13.8	14.7	2.3	0.3	70°	26°	0.0°
002	Vehicle	00:37	33.2	26.7	0.3	2.5	14.0	15.8	2.2	0.5	88°	93°	0.3°
003	Collie	00:33	33.3	24.3	...	...	16.5	17.1	0.1	0.8	92°	172°	0.0°
Basic parameters identifying the vehicle and time step.			Coordinates in both global (x,y) and road-relative (s,n) frames.				Velocities and accelerations aligned along the road.				Heading and heading rates (included to avoid recalculation).		



KEY FEATURES FOR RULE ENGINES:

Pre-calculated values | Standardized structure | Direct filtering support
No recalculation required | Enables lane-change rules



Summary

- Approach-aligned (s, n) coordinates separate forward from sideways motion
- Smoothed 30 Hz data plus central and second differences give v, a, ψ and ω
- Lateral movement is flagged by $|v_n|$, |ψ| and a minimum duration
- Relative Δs, Δn and Δψ support lane filtering and clearance analysis
- Output feeds the event definition and calibration of lane-filtering rules

THANK YOU

Presented By:
Saurav Barua

Date:
November 11, 2026